

SAADHAK

Most AI trading agents are confident. This one is calibrated.

An autonomous, defined-risk options agent on Alpaca that earns its position size by proving it knows what it knows.

TEAM SAADHAKA

lablab.ai x Alpaca AI Trading Agents Hackathon · paper account PA36JVGLDYR4

The problem

Confidence is not accuracy, and nothing usually checks.

A language model will tell you it is 90% sure.

Nothing scores that claim.

Nothing changes if it turns out to be wrong.

So the number is decoration, and the size behind it is a guess.

Saadhak makes the claim falsifiable, scores it, and lets that score decide how much it may risk.

What it trades

Defined risk, every time.

Iron condors and vertical spreads: they profit when the market stays inside a range.

one day to expiry, on liquid index products

never naked, never a market order

the maximum loss is known before the order goes in

Three components

Niyama, the discipline. Deterministic code.

discovers the universe, searches 70 structures per name, ranks by expected value
the only thing that can create an order

Saadhaka, the practitioner. A language model over Alpaca's MCP server.

33 research tools, none of which mutate. May veto, cannot start a trade.

Saakshya, the witness. Scores every probability, reconciles through the Alpaca CLI.

Nineteen gates between a candidate and an order

Every one explains itself; every refusal is journalled with its reason.

market state hours, days to expiry, reconciled, kill switch

risk appetite whitelist, max loss, portfolio caps, loss halts, sizeable

measurement expectancy: TP at half credit vs a 2x stop = a 2-in-3 breakeven

quality defined risk, liquidity, delta data, idempotent id, limit only, exits

circuits earnings guard, regime breaker, correlation

What earns the size

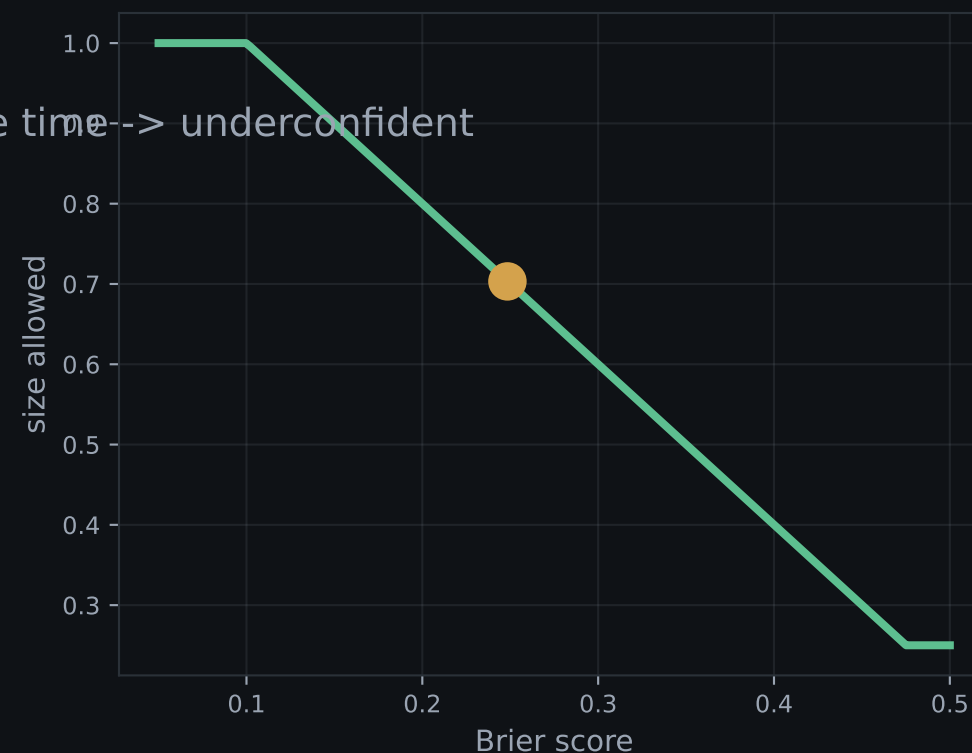
The mechanism the whole design rests on.

Every review states a probability.

The witness resolves it against what happened and scores it.

90% claimed and right half the time means the budget is cut

measured now: Brier 0.248 over 40; says 48%, right 57% of the time $\rightarrow$ underconfident



It knows when to stop

Two consecutive stop-outs block new entries in that name.

SPY and QQQ correlate at 0.91, so they count as one position, not two.

A 3% daily loss halts entries; 6% flattens everything.

exits keep running through a halt — it never traps a position

The stack

Every Alpaca surface, plus Featherless for the models.

Trading API	multi-leg limit orders, signed negative for credit; positions; history
Market Data	chains, bars, quotes, news — greeks solved locally at 0 DTE
MCP server	the practitioner's research, with the trading toolset withheld
CLI	reconciliation, kill switch, health checks, order dry-runs
Contracts	queried without an underlying, to discover what lists options
Featherless	open-weight models behind an OpenAI-compatible endpoint, serving the practitioner's review and the calibration study

The results, honestly

Four positions across two sessions. Three stopped out when the index trended.

the account finished at 99,212, *down*788

every loss capped at exactly one credit

no structure ever exceeded its stated maximum

Two sessions is not a track record. What this demonstrates is a mechanism.

What it rests on

Stated plainly, because it matters.

Alpaca's free data tier: IEX quotes, indicative option quotes, no OPRA, 15-minute SIP delay.

every decision records its data provenance and marks degraded inputs

a stock moved 8% between our last visible print and the truth

Calibration rests on tens of forecasts, not thousands.

Thank you

for taking the time to go through this agentic demo.

The model proposes. The discipline decides. A witness keeps score.

TEAM SAADHAKA

github.com/gachchami/saadhak
saadhak.streamlit.app

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